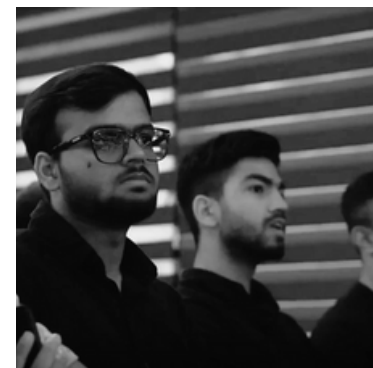
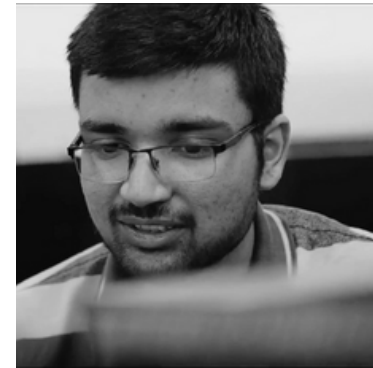


TechForChange

COHORT 4





Adarsh Goyal
Founder & CEO



Sneha Gupta
Co-Founder & COO



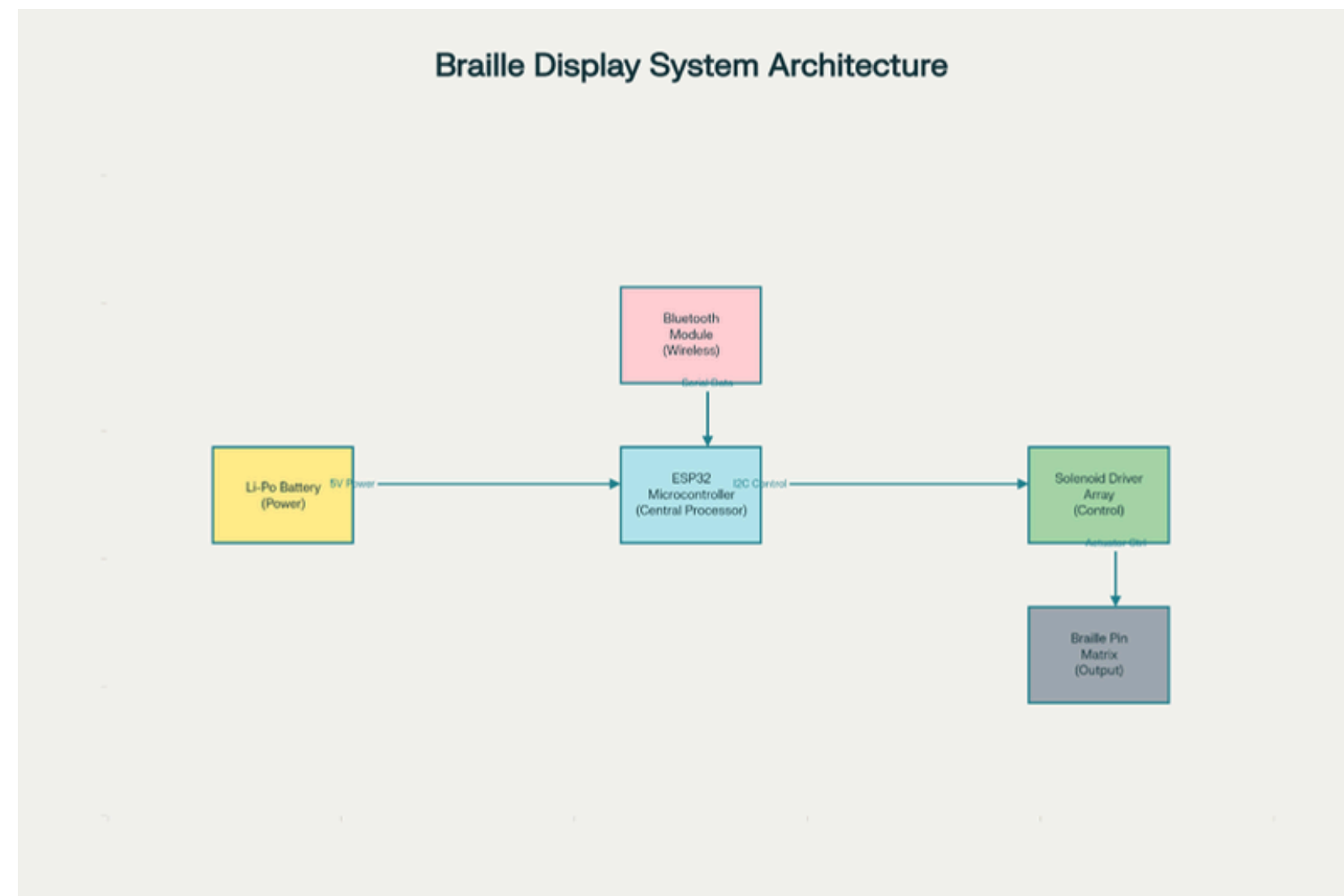
Naman Dadhich
Web Developer



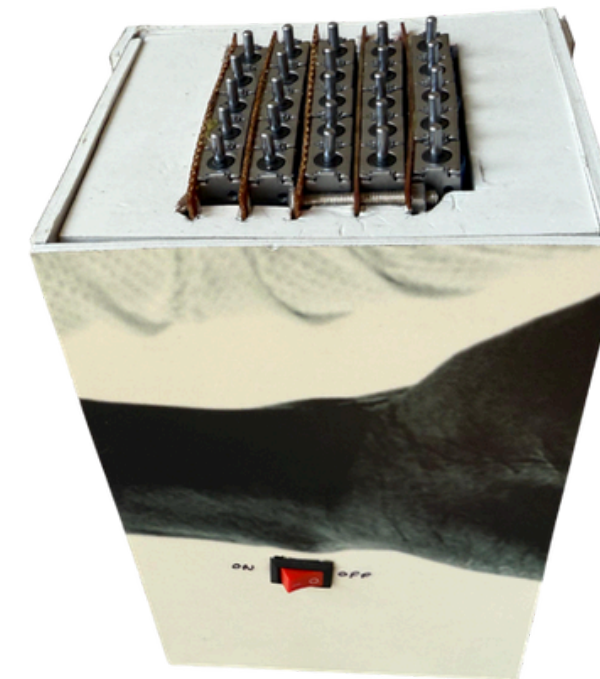
Sarthak Kakar
Web Designer

Our product uses intelligent software-hardware integration to empower visually challenged people in STEM Education

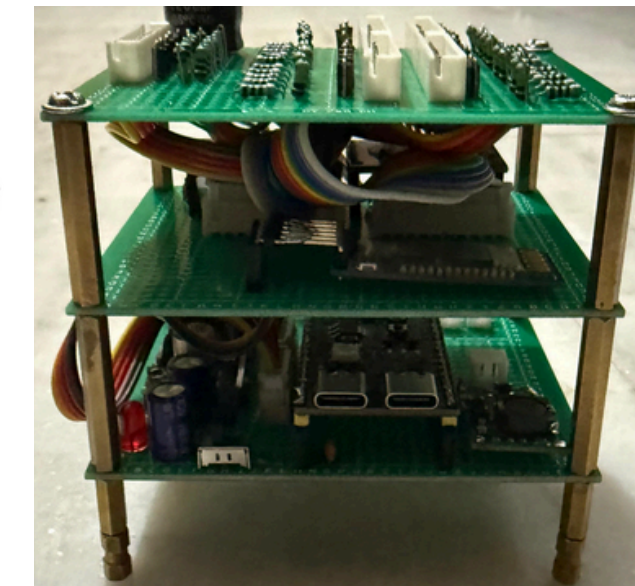
System Architecture



Hardware Prototype



Final Prototype



Internal Circuitry



Bluetooth-connected app drives a tactile pin-matrix via ESP32 + solenoid drivers.

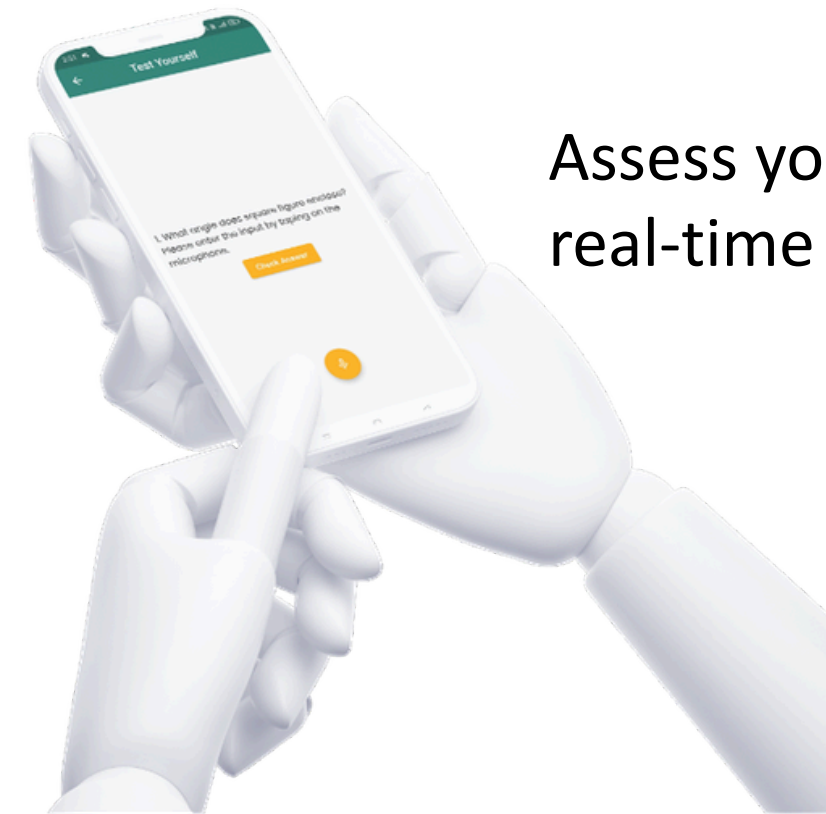


SOFTWARE SERVICES

- Interactive Learning Modules
- AI-Powered Feedback System
- Portable Tactile-Compatible Interface
- Multilingual Support



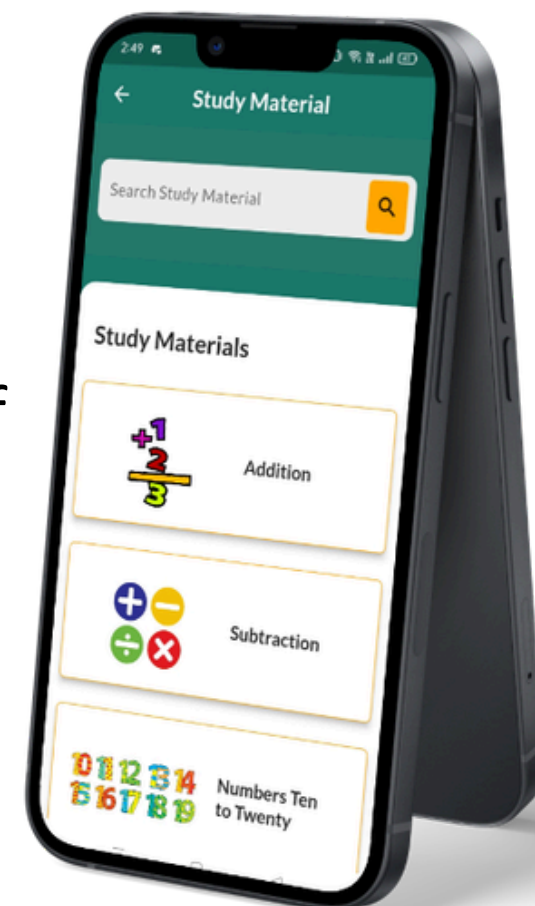
Use a Bluetooth-connected app and hardware to visually trace and comprehend figure geometry in real time



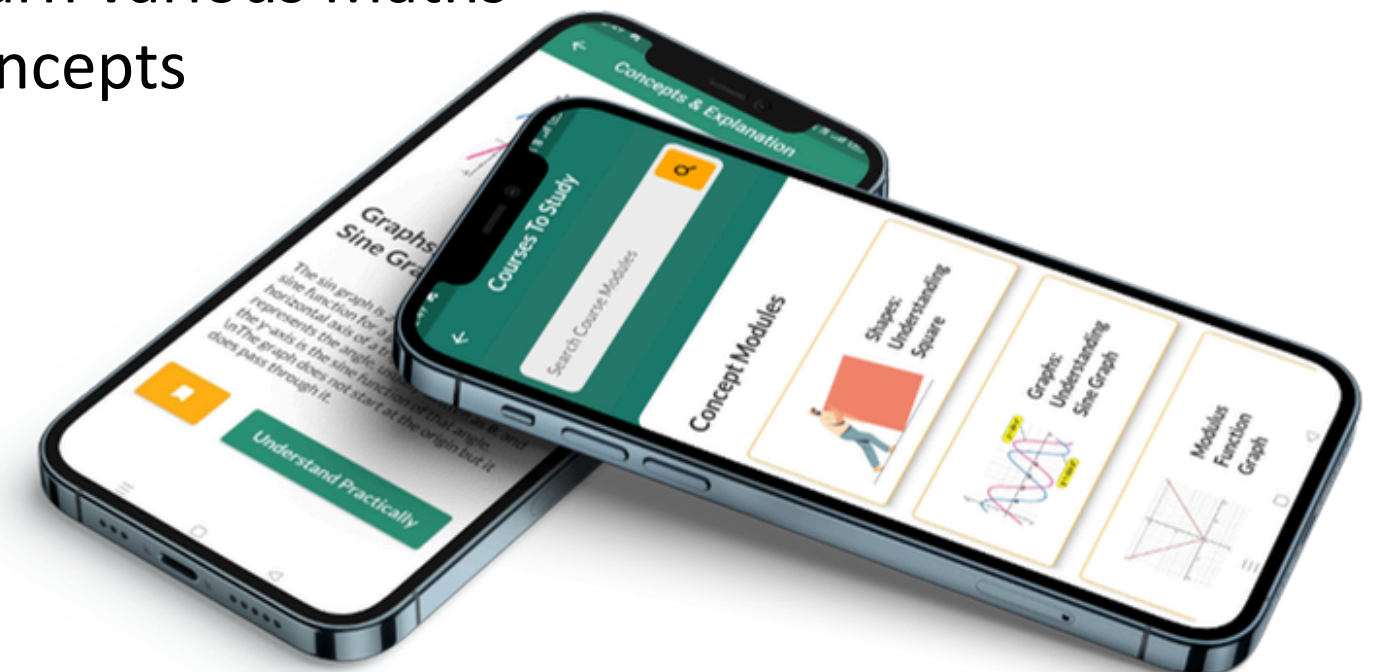
Assess your knowledge using real-time audio responses.

Key Features

Mobile app with voice read-aloud feature offering a diverse range of STEM study materials.

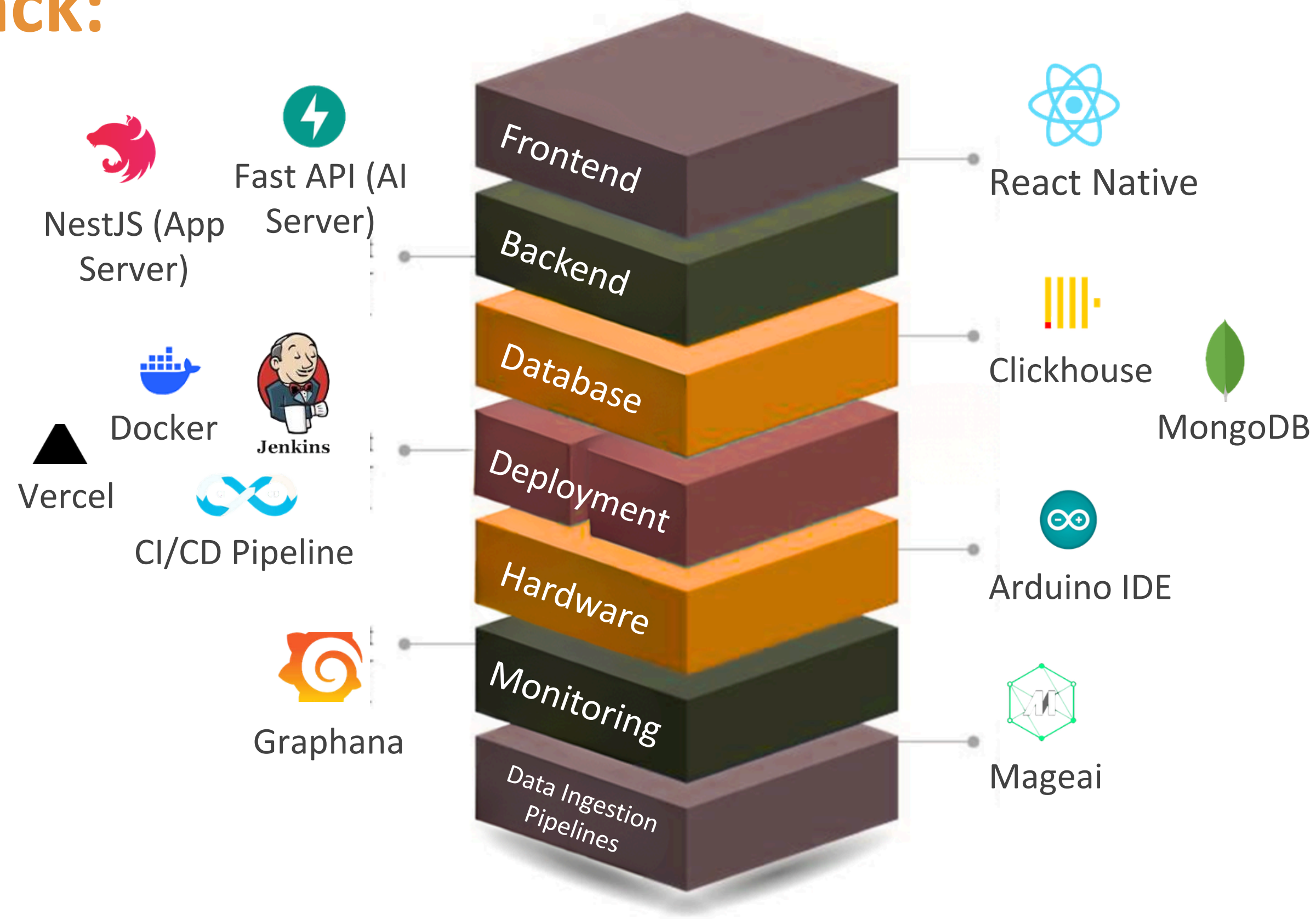


Explore and Learn various Maths and Science Concepts





Tech Stack:





Target Users:

We visited **Shri Sai Nethraheen Sanstha** and **Blind Person Association** in Uttam Nagar and we tested our product on visually impaired individuals.

Feedbacks Received Based on Interviews and Testing Sessions:

Right now we mostly rely on audiobooks, but they don't help us feel shapes. Geometry books and physical models are also heavy and hard to carry. Your device solves this problem because we can feel different shapes on a single device, and it is much easier to carry.

Rajesh

This product helps us imagine shapes in our mind. For ex- it helps us understand how A looks, how a triangle feels, or what other figures are like. For blind students, it is very helpful because it allows us to connect more with what we are studying. With this device, understanding shapes and figures becomes much easier.

Krishna

The device feels a bit large right now. If the size could be reduced, it would be better because then we could feel the whole image using just our palm. That would make it more comfortable and easier to use.

Rajesh

15+ Users Tested



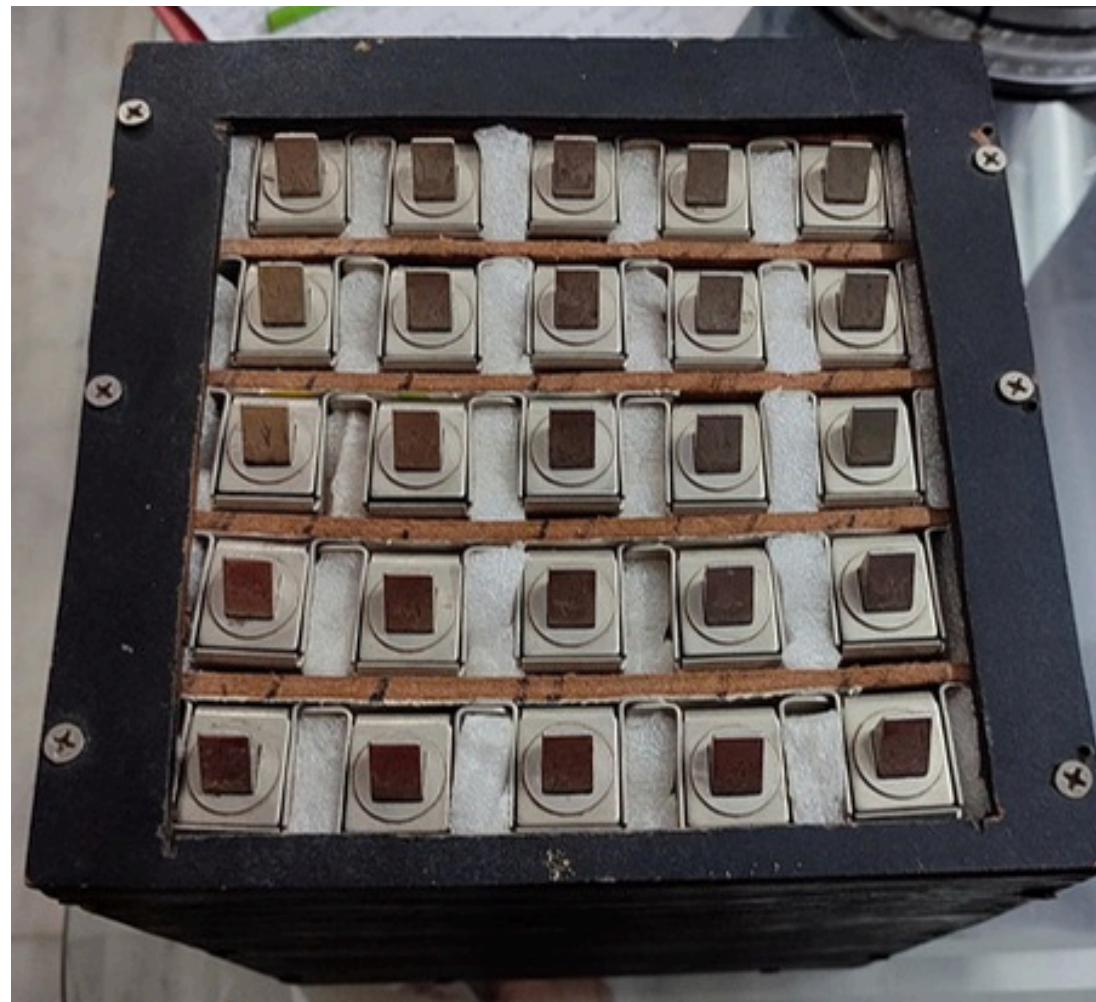
User Validation & Confirmation

Click Here - https://www.youtube.com/watch?v=P_1gWaKteHY

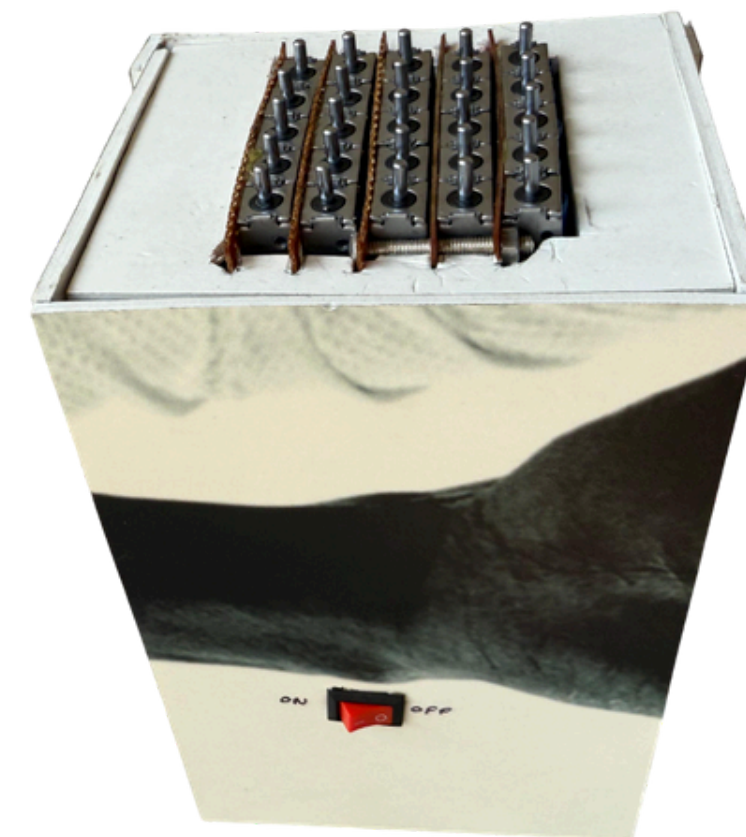
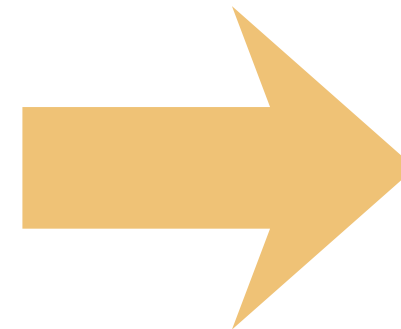




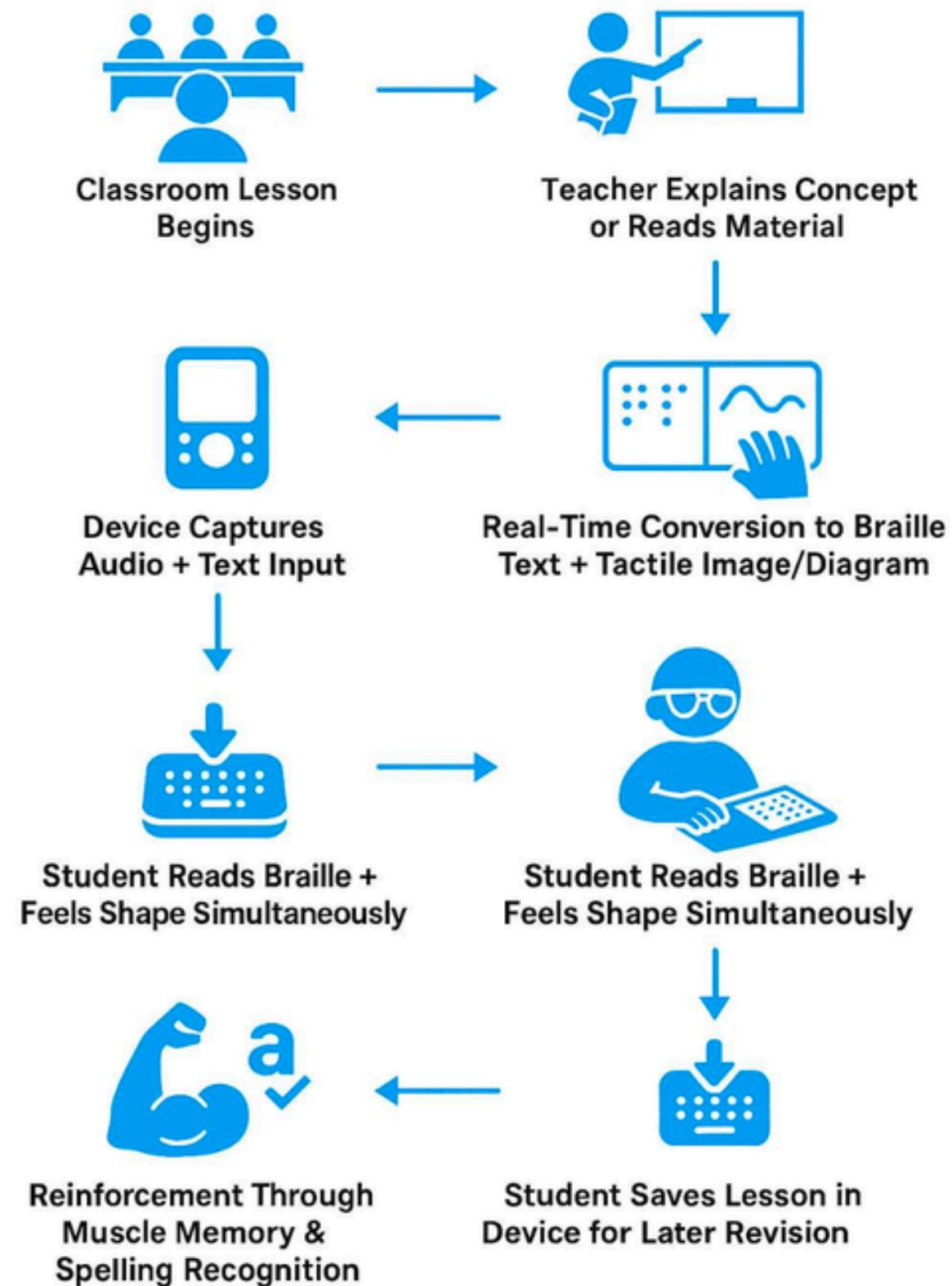
Based on the feedback we received, we redesigned the hardware and made it smaller in size. This allows users to easily feel the images using their palm, and the device can now comfortably fit in the hand.



Before



After



**Better Understanding → Improved Confidence
→ Higher Academic Performance**

Friction Points Identified:

- In app, we are currently using talkback feature, but we need to implement voice controlled application.
- In hardware, a lot of heat is produced due to solenoids, so we need to reduce the heat that is being produced.



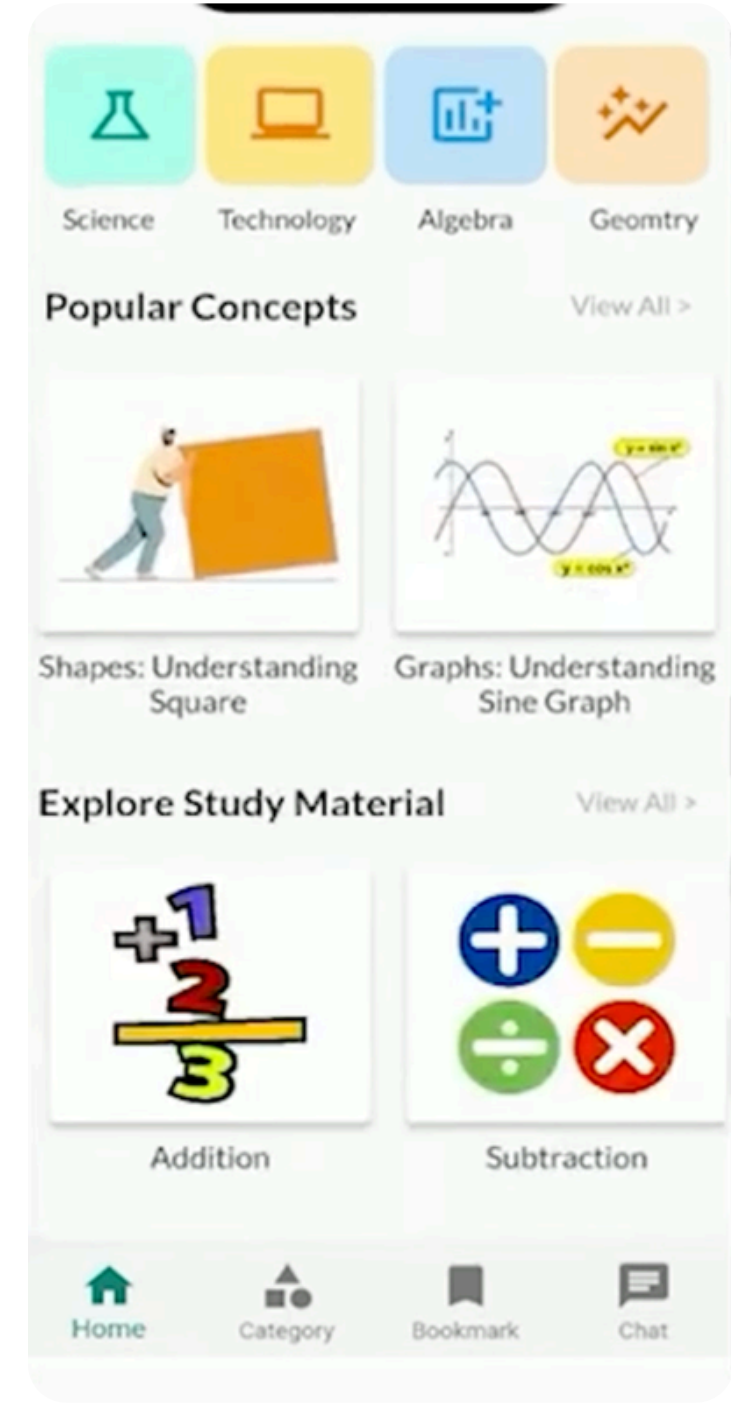
Screens



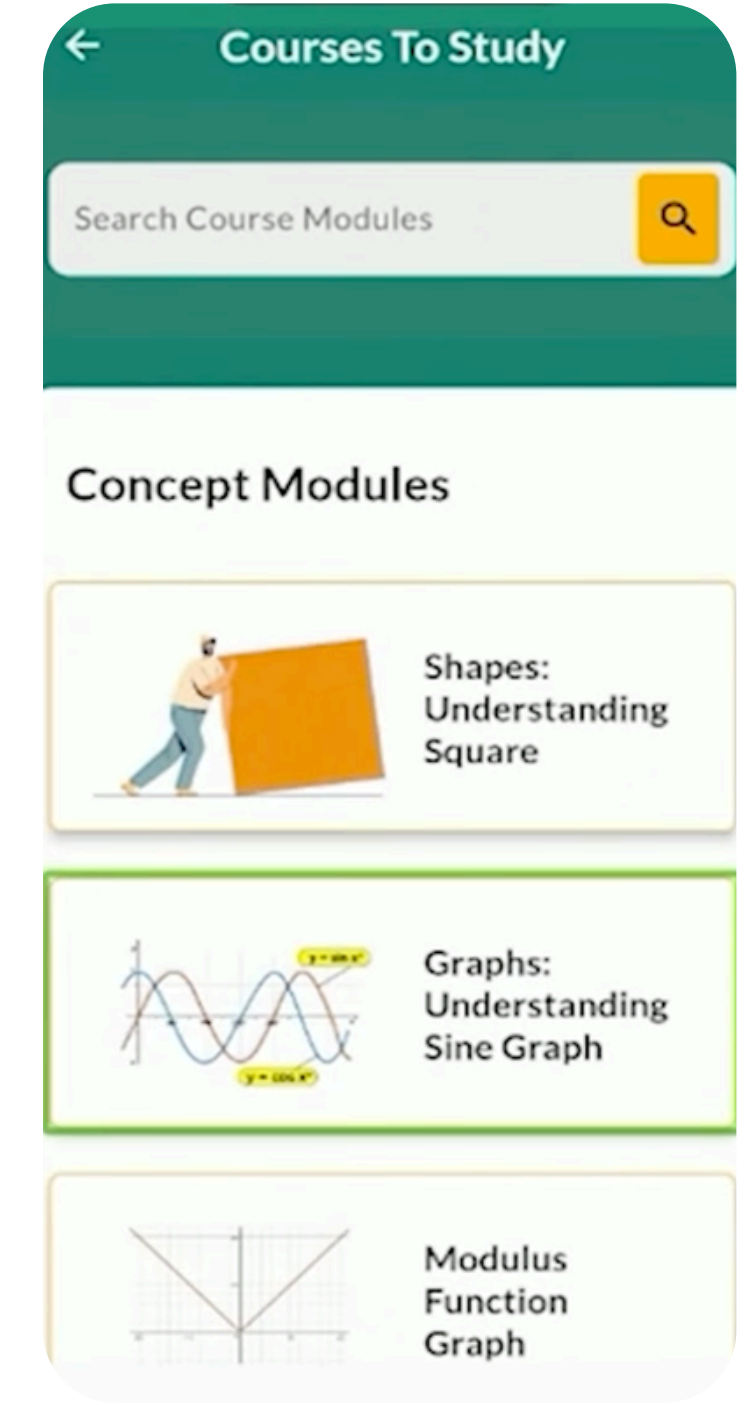
Starting with the app



Available options to learn, practice and test



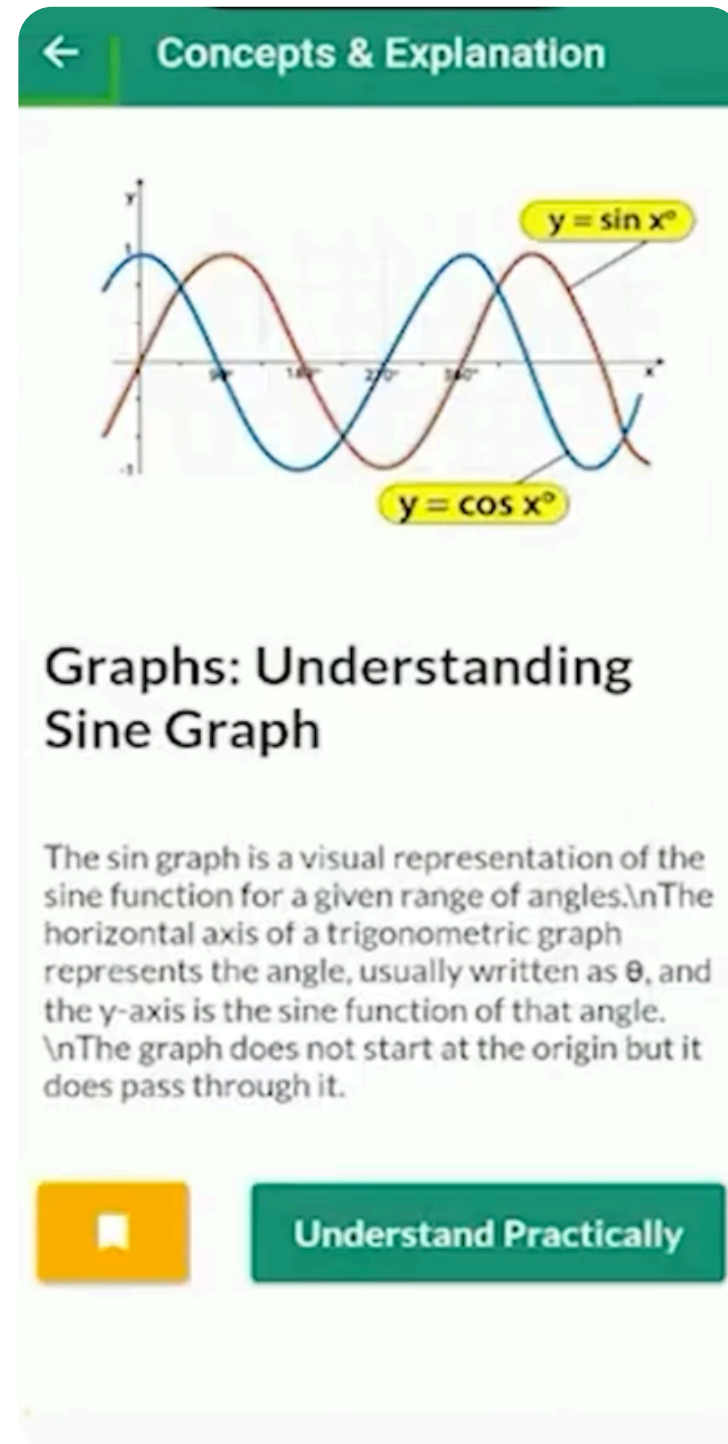
Offering a diverse range of STEM study materials.



Explore and Learn various Maths and Science Concepts



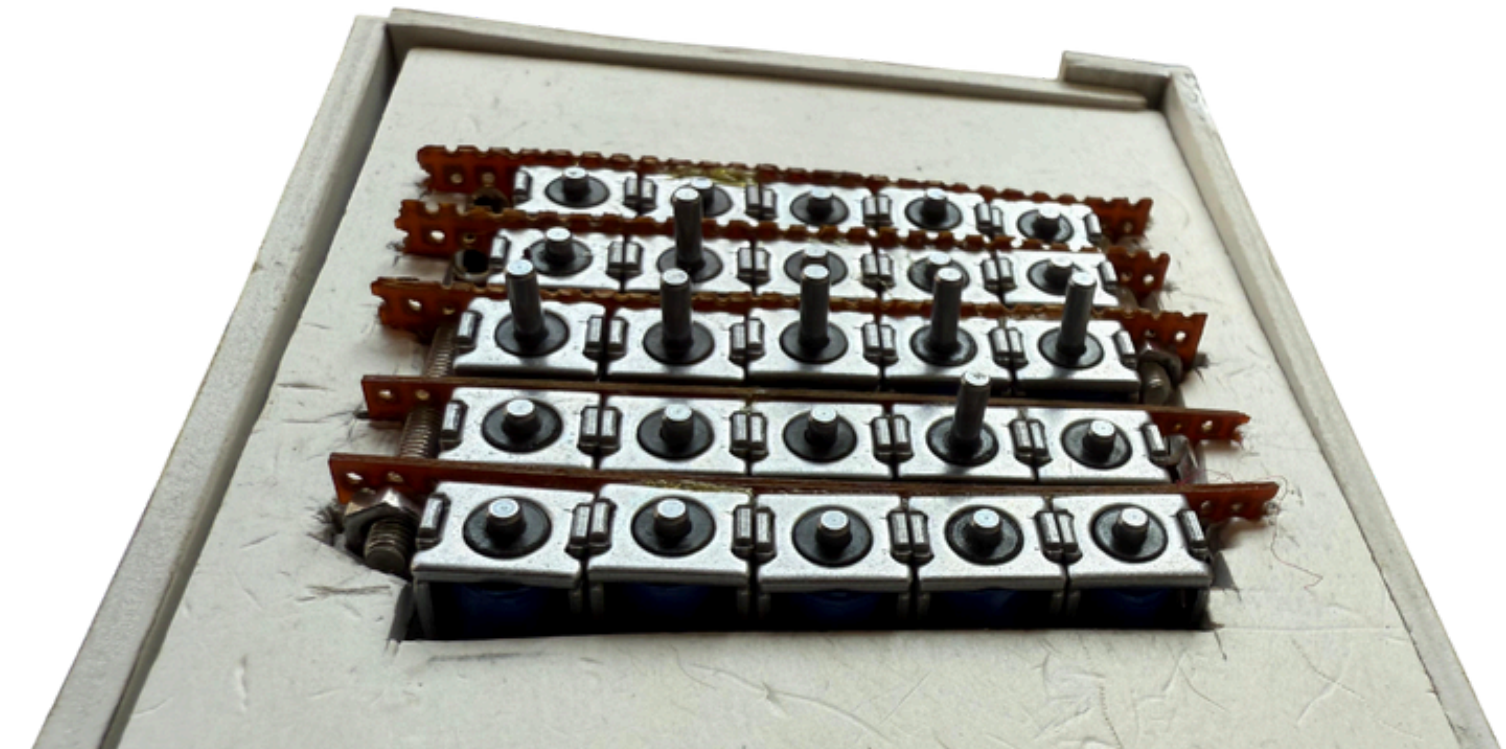
Screens



Understanding Sine Graph



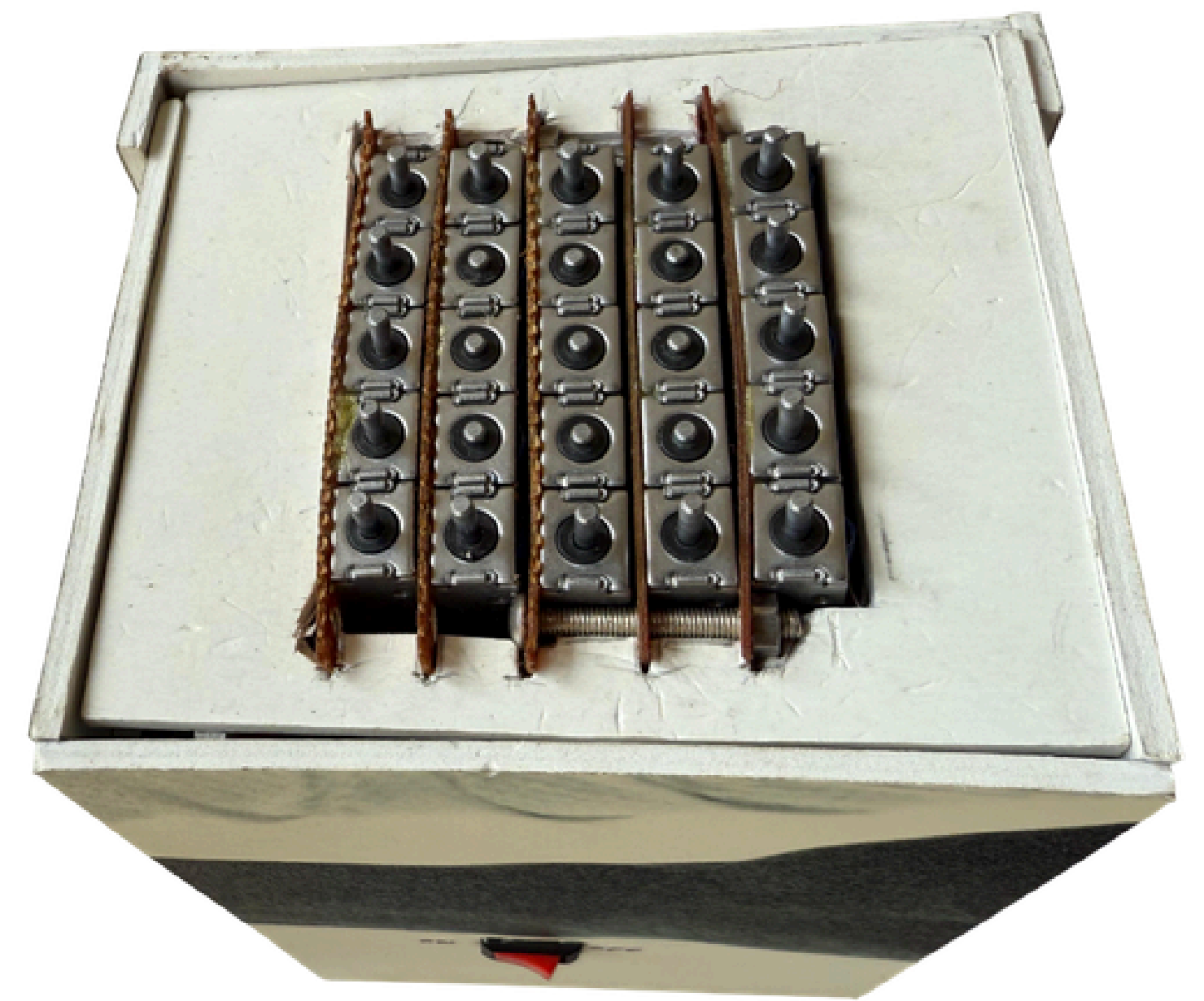
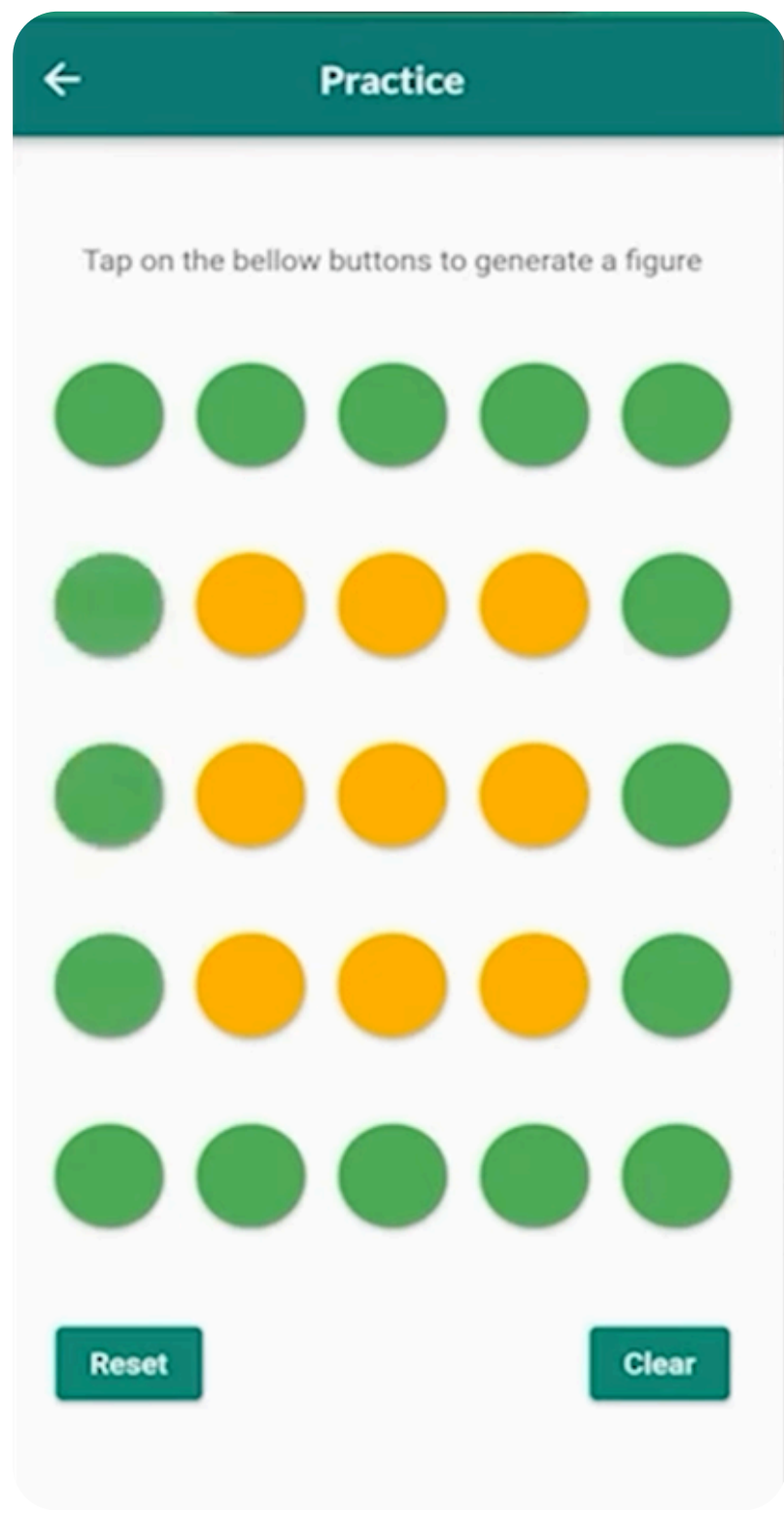
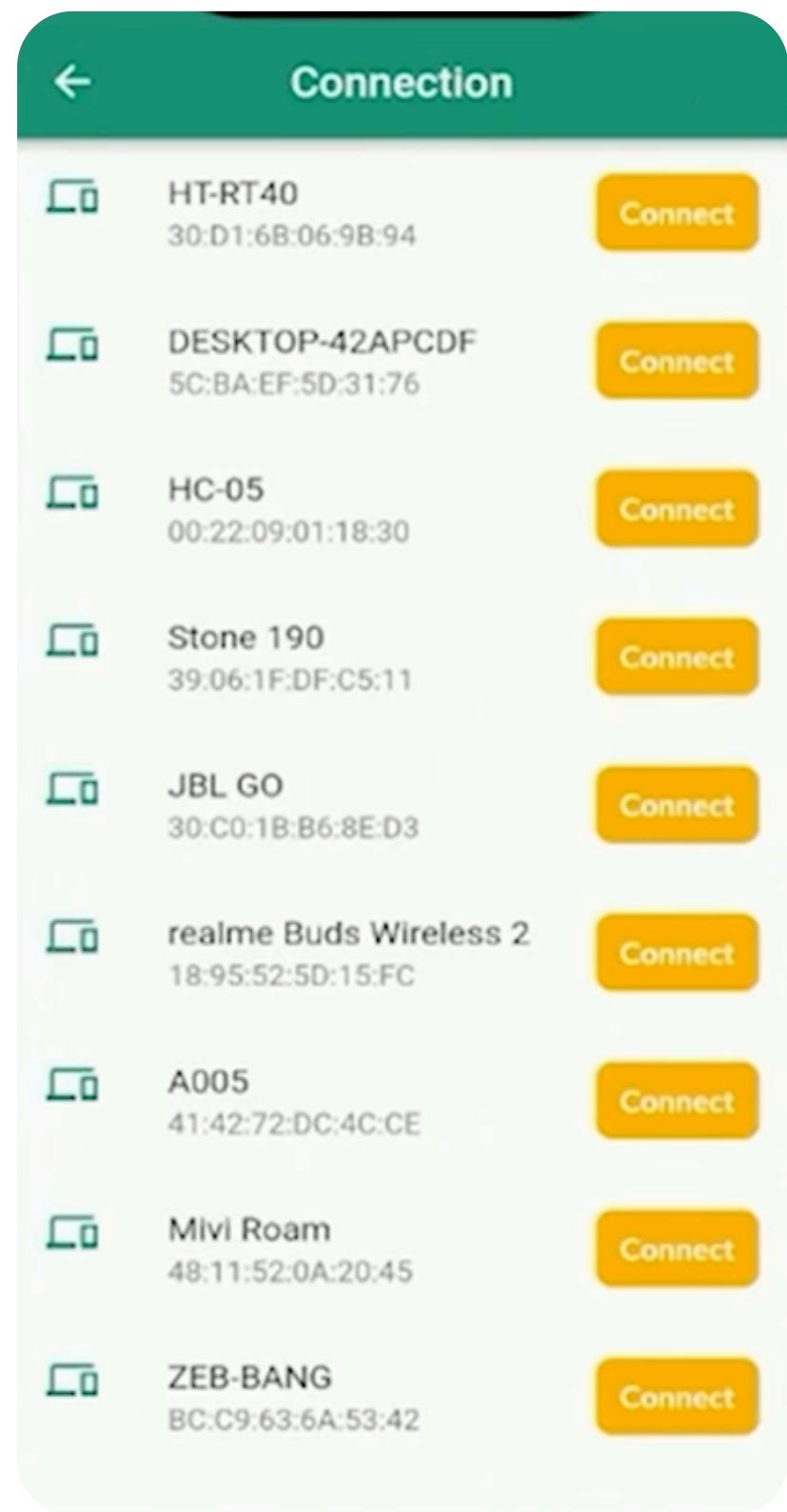
Bluetooth enable option for connecting to hardware



Hardware showing the practical representation of Sine Graph



Screens



Connecting Bluetooth

Drawing square on app

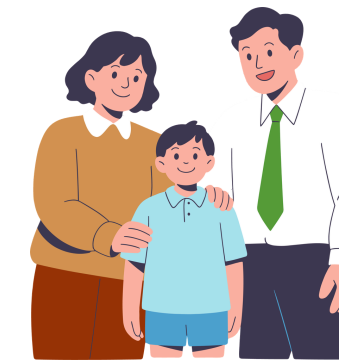
Hardware showing
square figure



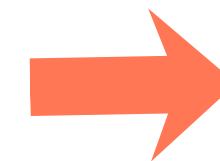
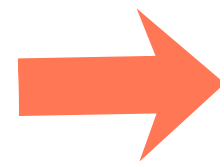
Who is Benefited?

Primary - Visually impaired / blind students (especially in under-resourced schools)

Secondary - Teachers, special educators, schools, parents/caregivers, NGOs working in disability inclusion



Real World Use Case:

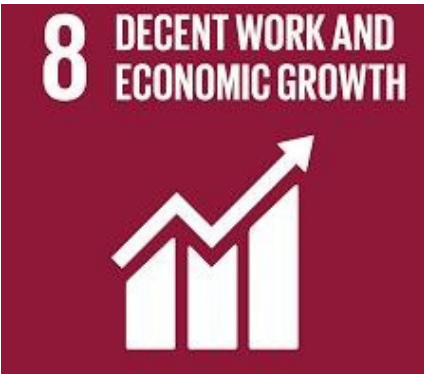


Teacher introduces topic (e.g., triangles / graphs / coordinate geometry)

Student uses the tool to feel the diagram (tactile output) + hear guidance (audio)

Student solves practice questions independently, gets feedback



SDGs	Direct Impact	Indirect Impact
	Accessible STEM via tactile + audio learning; improves concept understanding and confidence	Teacher enablement + curriculum integration for inclusive STEM classrooms and assessments.
	STEM readiness → pathways to higher studies, skills development, and employability.	Long-term economic inclusion by reducing dependency and widening career opportunities.
	Affordable assistive-tech learning infrastructure (device + app ecosystem) for learners.	Modular design supports scalable manufacturing, local repair, and sustainable deployment.
	Cuts disability-based learning barriers; supports equal participation without constant assistance.	Scales access through schools/NGOs to underserved communities and low-income learners.

Estimated Scale



Pilot target
(first deployment)

3–5 schools /
50–150
students

Post-pilot
(after iteration)

10–20 schools
/ 300–800
students

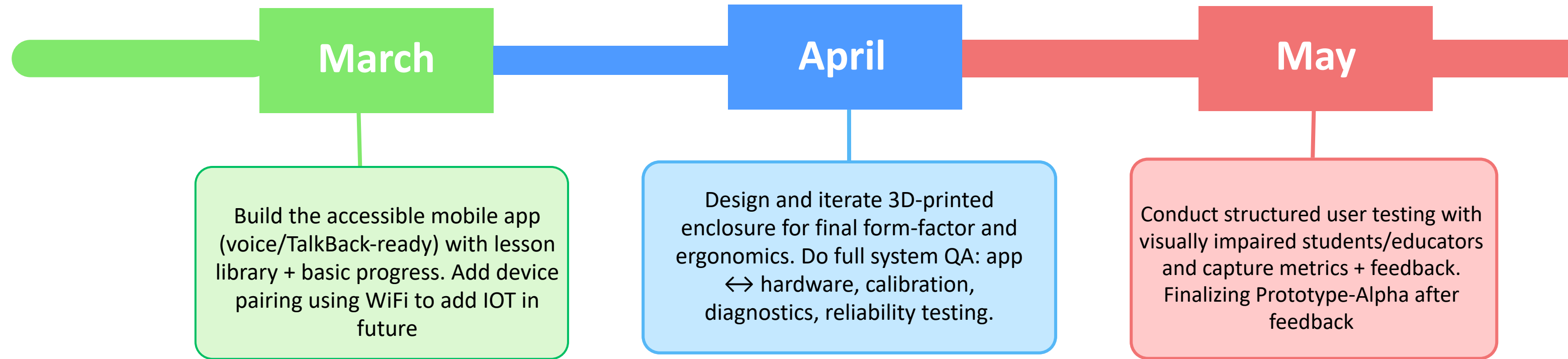
Measurable outcomes



- ✔ **Learning gain:** pre/post concept test improvement (%)
- ✔ **Independence:** % tasks completed without assistance
- ✔ **Time efficiency:** time to complete diagram-based problems
- ✔ **Teacher adoption:** weekly class sessions using tool
- ✔ **Retention rate:** attendance or participation proxy
- ✔ **Cost proxy:** reduction in material dependence per learner



3 months roadmap:



Deployment plan (how it scales)

- ✓ **Manufacturing:** small batch production → QA checklist
- ✓ **Training plan:** train-the-trainer model (NGO/lead teachers)
- ✓ **Support plan:** hotline/WhatsApp support + maintenance flow
- ✓ **Data/reporting:** monthly KPI review + impact reporting pack

Key milestones

- ✓ **M1:** Pilot prototype + training kit ready
- ✓ **M2:** First classroom pilot live
- ✓ **M3:** KPI report + v2 iteration shipped
- ✓ **M4:** Expanded deployment cohort



Engagement with Program (Mentoring, Events, Replenishment)

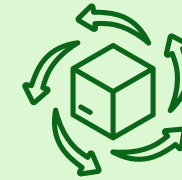
We received 16 hours of mentorship, and their valuable feedback helped us develop the hardware smoothly. Their relentless support during the development of the app was also greatly appreciated.

Mentorship



We requested several replenishment supports for hardware development, and all were successfully approved, enabling us to build the hardware efficiently.

Replenishment



Events & Workshops



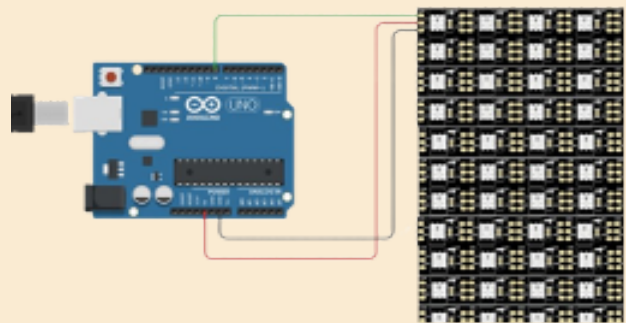
We have attended all the events that have taken place so far. Additionally, out of the 16 workshops conducted, we have attended 11.



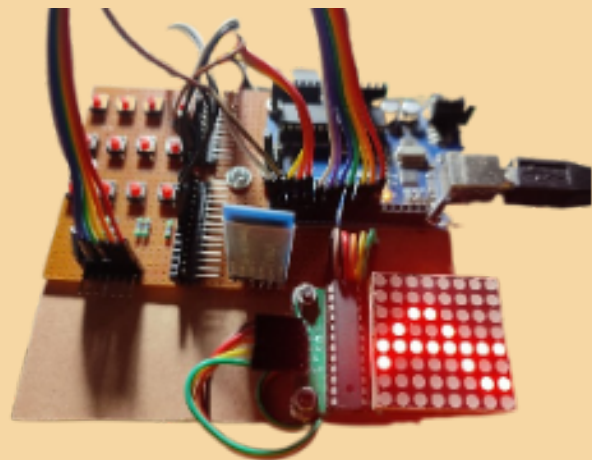


Evolution of Idea

- Researched the problem, its solution and market need.
- Developed a concept feasibility model.



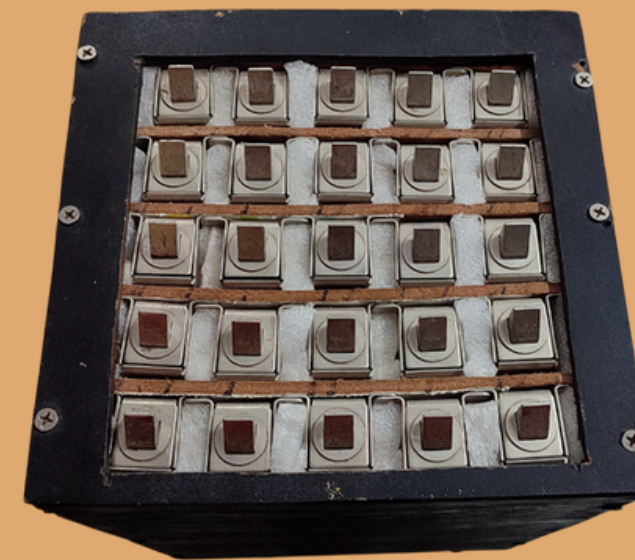
- Developed a basic UI functional android application.
- Implemented the feasibility model and built a functional LED prototype.



- Improved the user experience of the mobile application.
- Built the fully-functional market ready hardware.



- This is still work in-progress.
- Our team is working to achieve this hardware design appearance.



- Mobile Application is ready and tested.
- The hardware prototype is fully ready and meets the customer's expectations, being compact and easy to handle.





Challenges Faced:



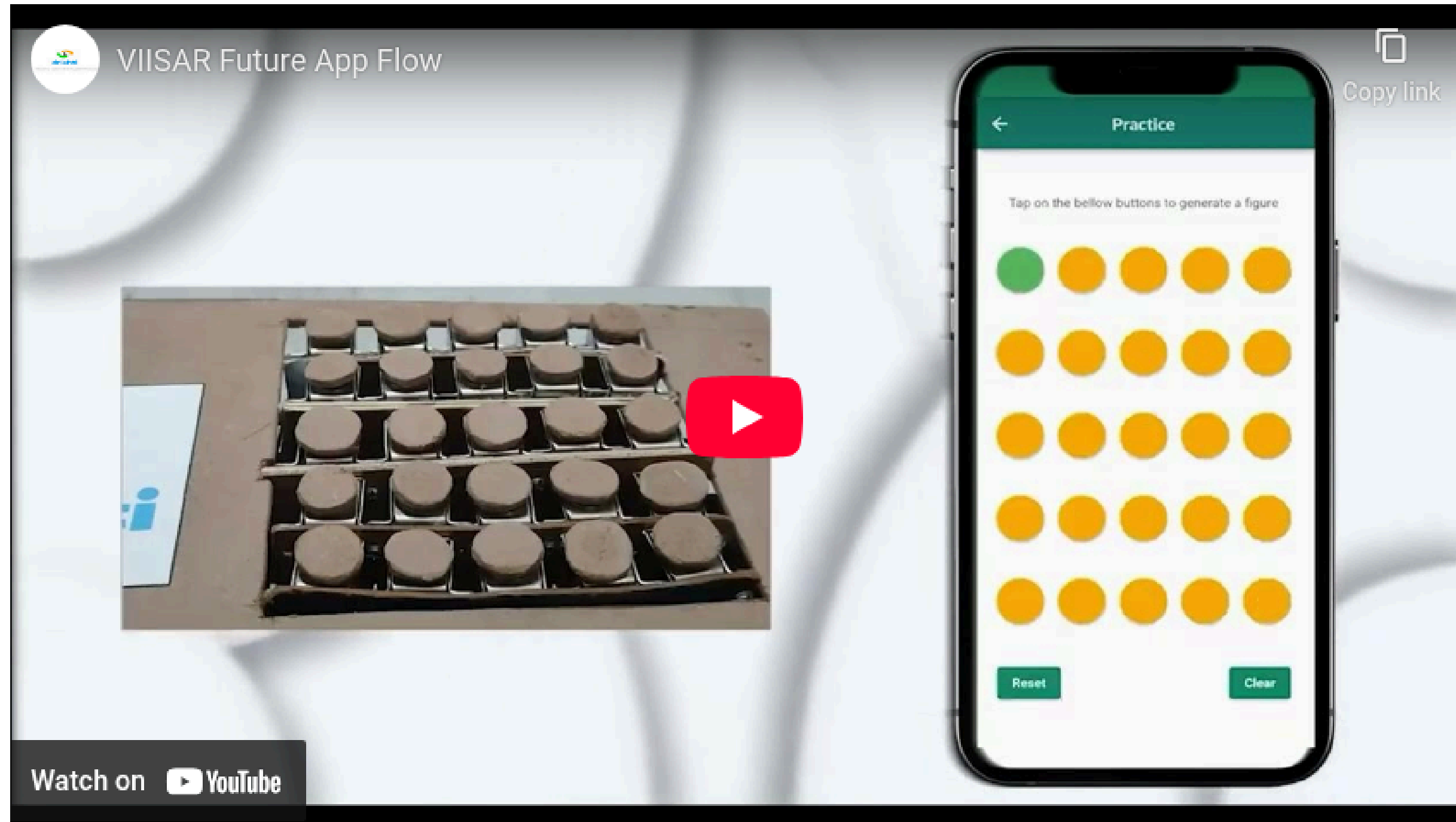
01. To maintain and build bluetooth connectivity feature in app,

02. To reduce the size of the hardware

03. To reduce the current consumption and heat dessionation in hardware



Video of the MVP to be attached here



[Click Here - https://youtu.be/COelZeo9sno](https://youtu.be/COelZeo9sno)

Thank You

